Disease Classification:
Knowledge Graphs & GCNs

David Sherman & Janhavi Subedarpage

Goal & Background

- Goal: To develop a model, using knowledge graphs and graph convolutional networks (GCNs), that can classify diseases based on provided patient symptoms
- Model applications
    - Classify illnesses and treatments more efficiently
    - Reduce room for error
    - Useful for high risk diseases that require immediate care
    - Knowledge graphs offer a structured means of capturing relationships between entities

Datasets

Dataset 1 - Patient Narratives

Disease	Patient Symptoms
Diabetes	I have a dry mouth and throat. I also have been experiencing an increased appetite and hunger. However, I do feel very tired at times

Dataset 2 - Knowledge Graphs

Disease	Symptom 1	Symptom 2	Symptom 3
GERD	Stomach pain	Tongue ulcers	Cough
Diabetes	Fatigue	Weight loss	Irregular sugar level

Models/Methods Used

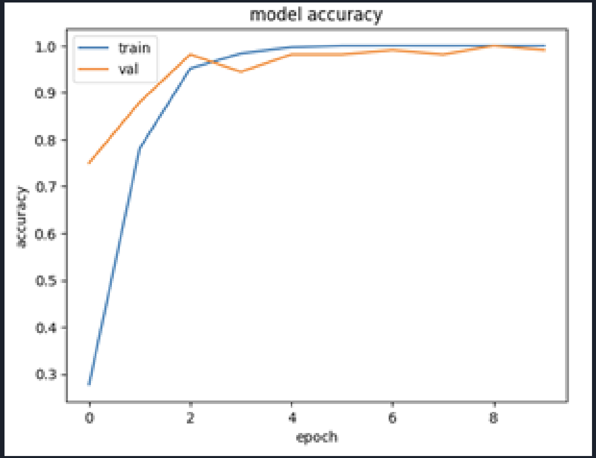
- Baseline Model
  - Used BERT for multi-class text classification
- Knowledge Graphs
  - Used to depict relationships between diseases and symptoms
- Graph Convolutional Networks
  - Used to understand and learn from data organized in a graph structure, making them particularly useful for tasks where relationships and connections between entities are crucial

Methodology

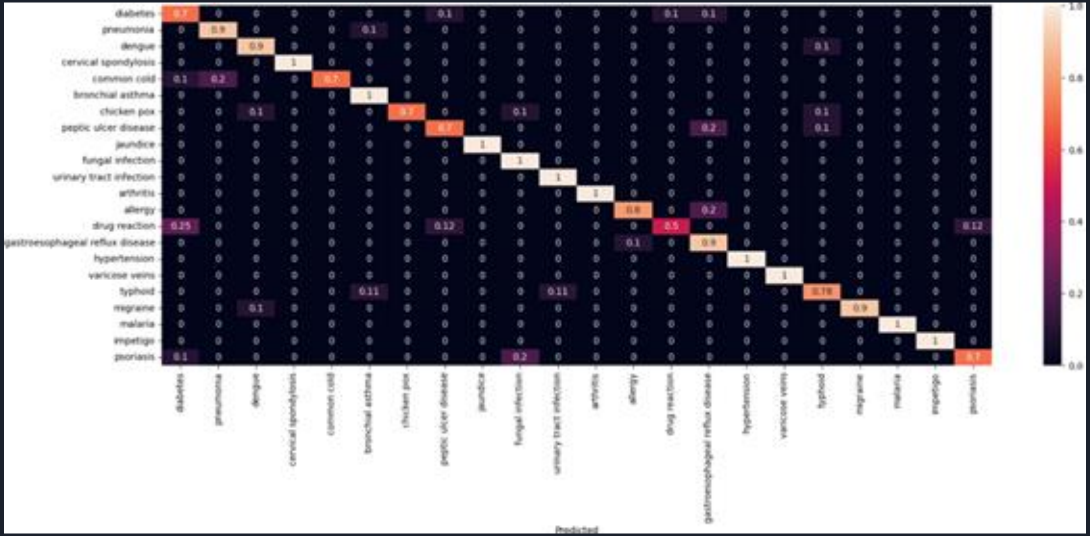
1. Trained BERT model by using splits of 80/10/10% for training, validation, and test datasets, respectively
   a. Implemented ‘bert-base-cased’ tokenizer to tokenize and encode sentences (patient narratives)
2. Constructed knowledge graphs by mapping all diseases and associated symptoms as source and target nodes
3. Built the graph convolutional network, which involved text cleaning, making the knowledge graphs, then training the GCN

Key Figures

Baseline Model Performance

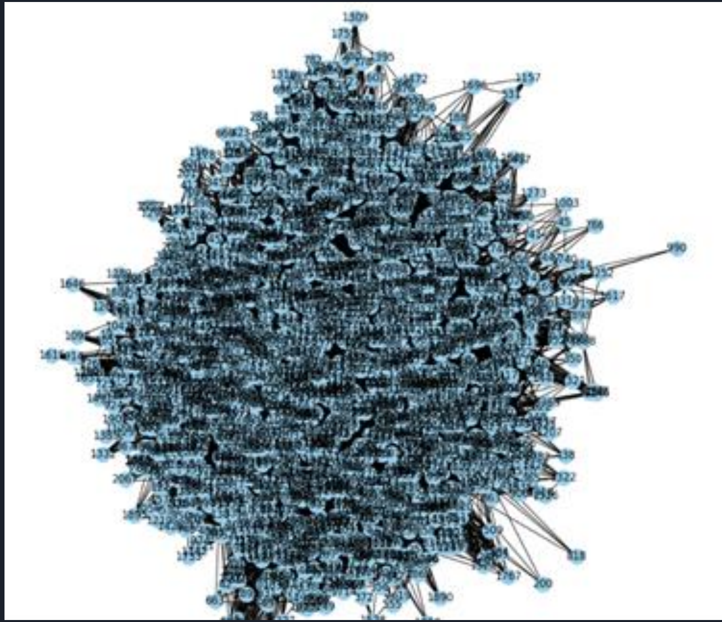


Confusion Matrix - Similar Diseases

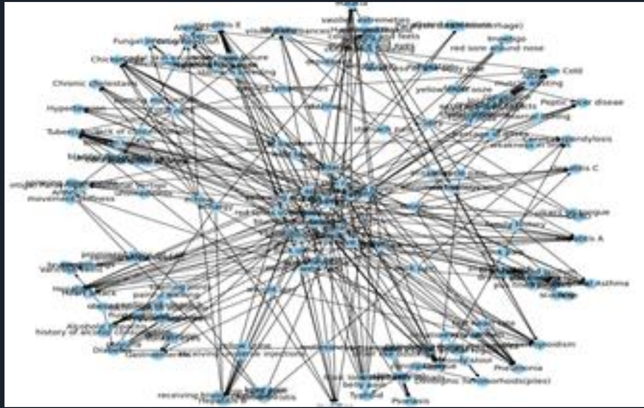


Key Figures

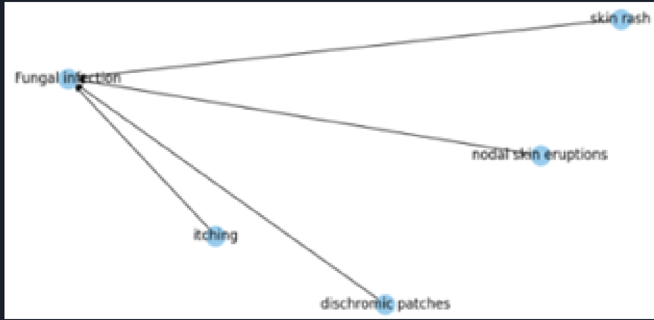
Knowledge Graph with TF-IDF Vectors



Knowledge Graph - All Diseases



Knowledge Graph - Single Disease



Findings

Model	Loss	Precision	Recall	Macro F1
BERT (Training)	0.505	0.87	0.87	0.87
Text GCN (Test)	0.202	0.94	0.93	0.93

- BERT model was prone to misclassification (i.e. confused common cold for pneumonia)
- GCN model yielded more accurate results, as it did a better job capturing semantic relationships between diseases and symptoms; also involved:
  - Meticulous cleaning of patient narrative text
  - Introducing a knowledge graph based on TF-IDF vectors
  - Pointwise mutual information (PMI) calculations

Conclusion & Next Steps

- To enhance the capture of context and semantic relationships in textual data, we can consider integrating attention layers into the GCN or combining them with pre-trained transformers like BERT
- If allotted more time, we would have liked to build a chatbot capable of processing user symptoms and returning classified illnesses and suggested remedies

References

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Questions?